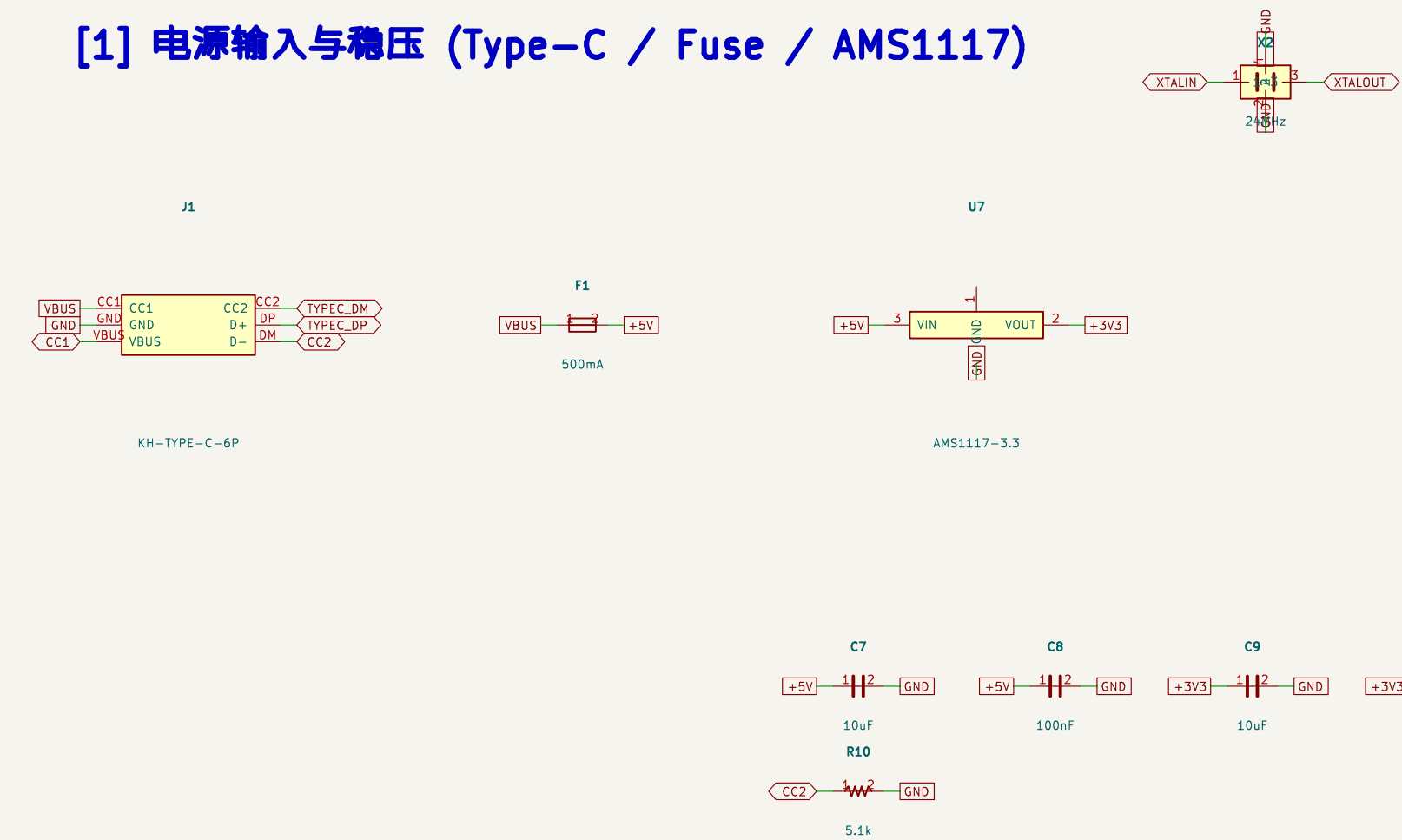


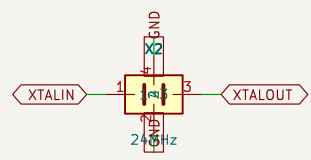
# MBDSDR ai-sdr Mini 原理图 v0.7.1

AI-Defined Radio | 80x60mm 2-layer | 62 components / global-label nets | GPL-3.0 | BI4MIB

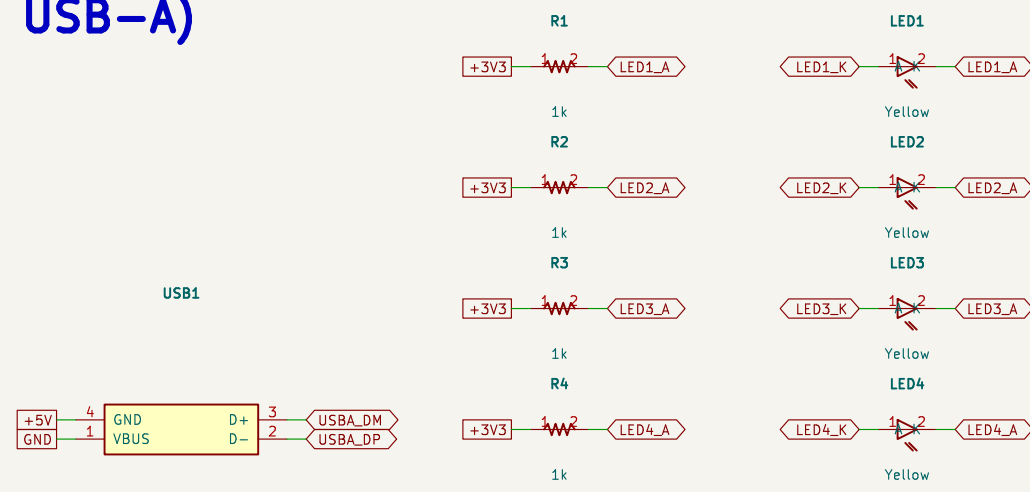
## [1] 电源输入与稳压 (Type-C / Fuse / AMS1117)



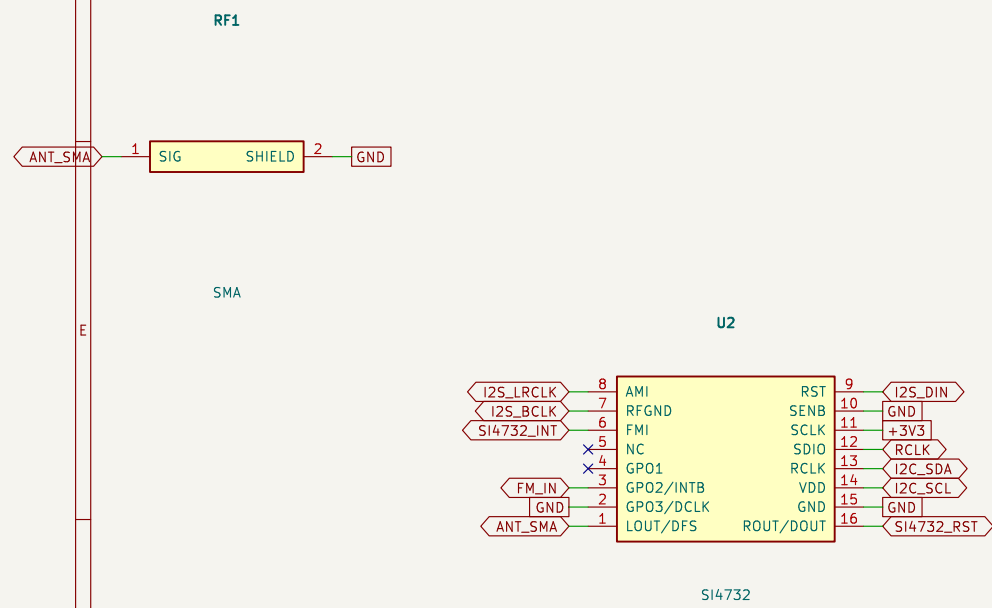
## [2] USB2514B Hub + 24MHz (上行Type-C / 下行ESP32 & USB-A)



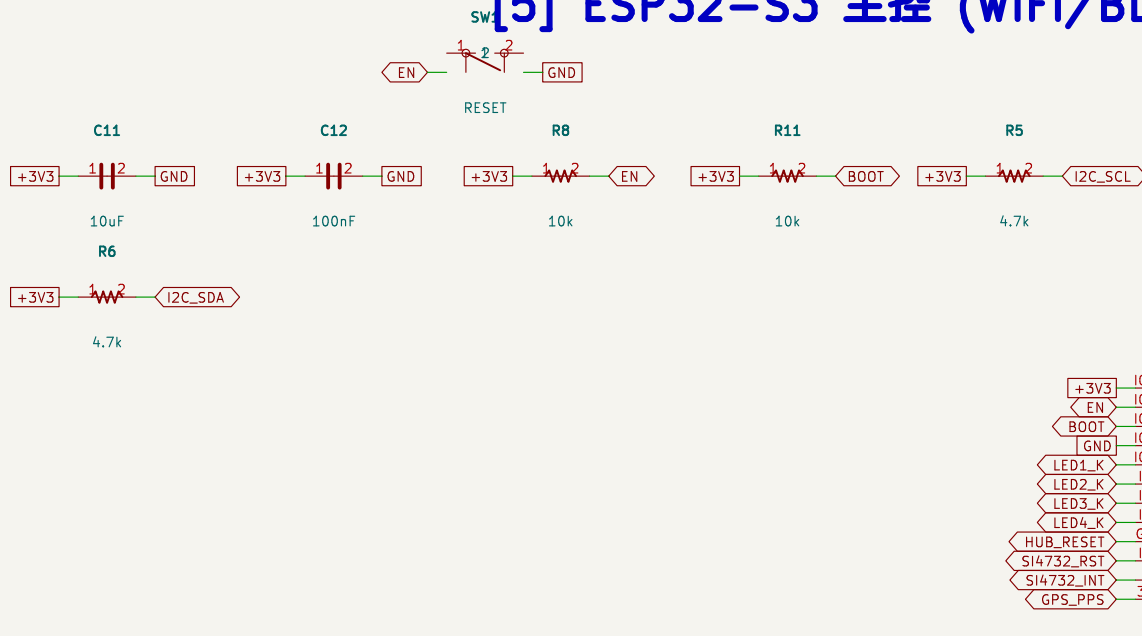
## [3] USB-A扩展口 / 状态LED



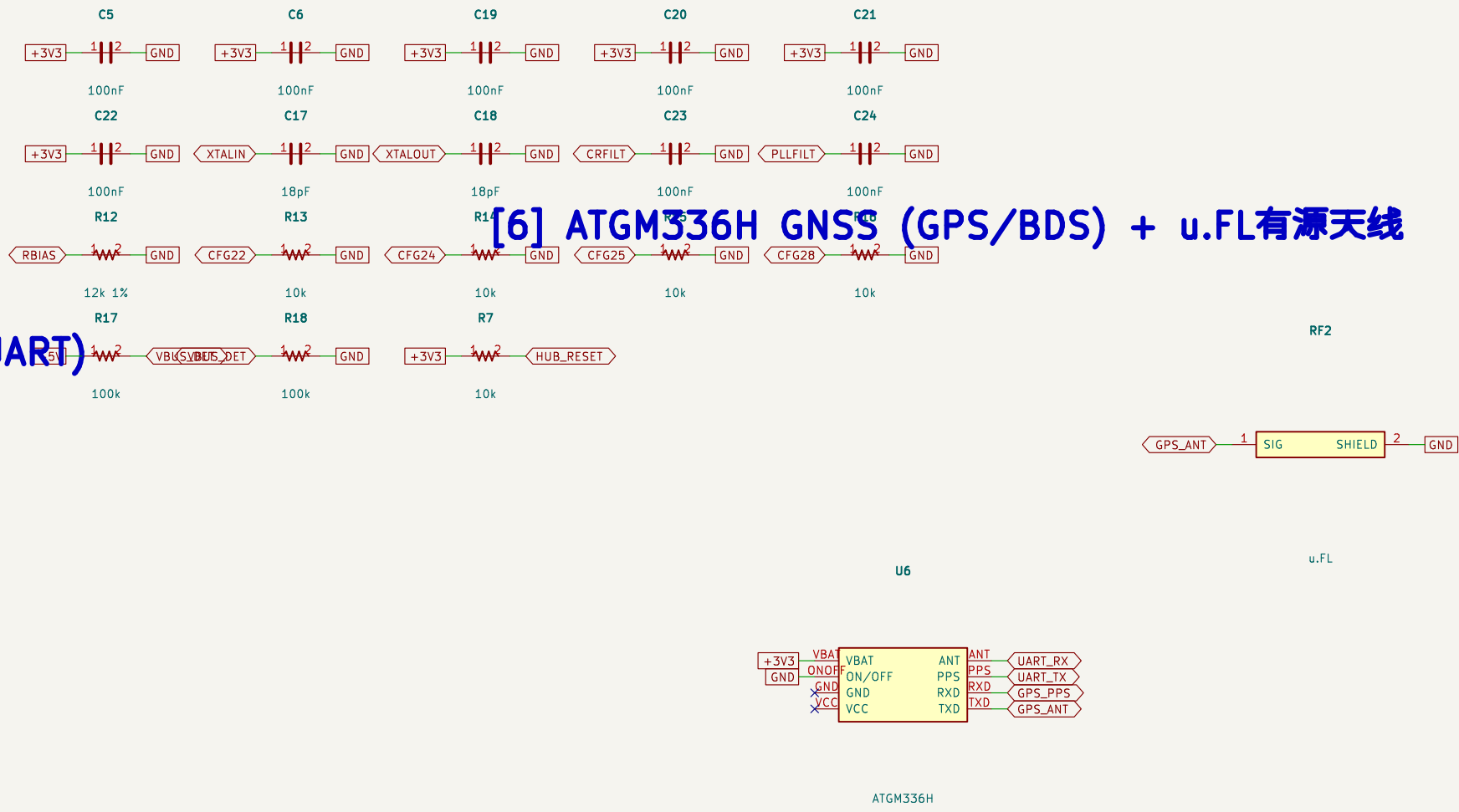
## [4] SI4732 广播接收前端 (AM/FM/SW + I2S) + SMA



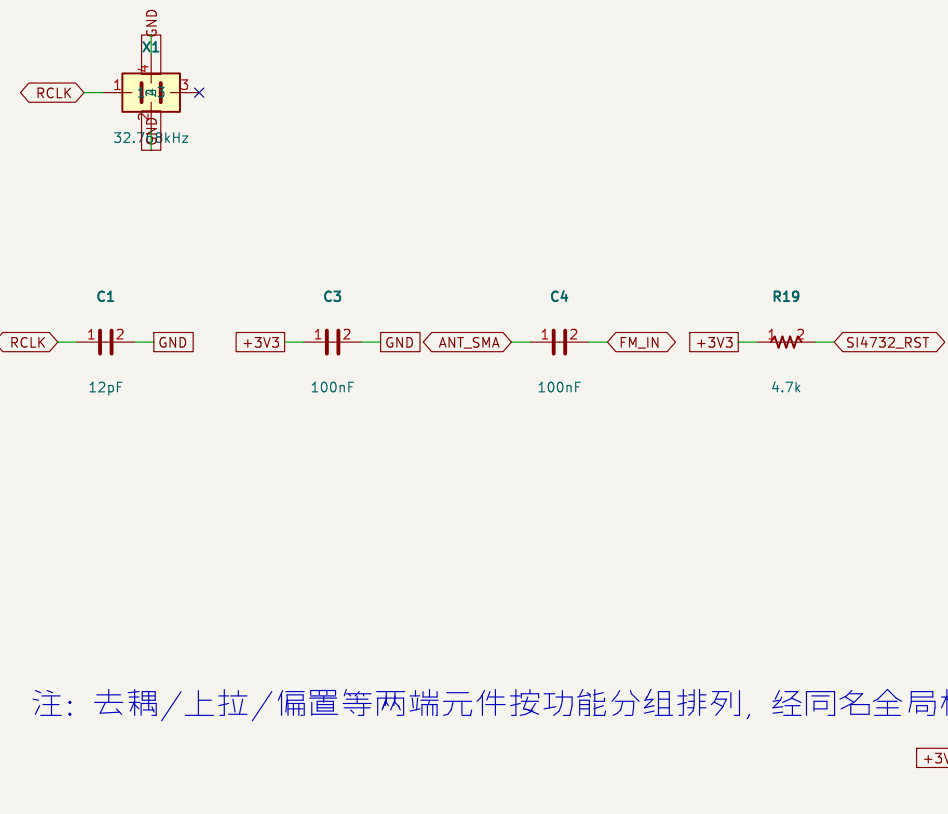
## [5] ESP32-S3 主控 (WiFi/BLE/USB/I2C/I2S/UART)



## [6] ATGM336H GNSS (GPS/BDS) + u.FL有源天线



## [7] 传感器 BMI260 IMU + TMAG5273 磁力计 (I2C)



注：去耦/上拉/偏置等两端元件按功能分组排列，经同名全局标签连网；投产时模组符号(ESP32/ATGM336H/BMI260)在立创/Altium替换为官方库封装，按GPIO/信号名网络连接。

SI4732 + ESP32-S3 + USB2514B + GNSS + IMU + Mag  
80x60mm 2-layer, JLCPCB SMT  
GPL-3.0 License  
Complete v0.7.1 schematic, global-label nets, auto-routable  
**MBDSDR Open Source Project (BI4MIB)**  
Sheet: /  
File: T5\_full.kicad\_sch  
**Title: MBDSDR ai-sdr Mini - AI Defined Radio Hardware**  
Size: A2 Date: 2026-09-16 Rev: v0.7.1  
KiCad E.D.A. kicad-cli 7.0.11-7.0.11-ubuntu22.04.1 Id: 1/1